Venkata Lakshmi Parimala Pasupuleti

Senior AI Engineer · Agentic AI · LLM Evaluation · AI Governance · Cost Management · Guardrails · Python · Azure · AWS

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SUMMARY

Senior AI engineer with 9+ years of software engineering experience, specializing in agentic AI platform design, LLM evaluation and governance, AI cost management, and production guardrails. Currently at Intuit integrating Anthropic Claude into production financial services: structured outputs, retrieval-grounded RAG pipelines, LLM-as-judge evaluation harnesses, guardrails with sensitive-data handling, and CI/CD deployment on AWS. Built **AgentProbe** - an enterprise-grade LLM evaluation and cost management platform: multi-provider routing across Anthropic, OpenAI, Azure OpenAI, AWS Bedrock, Groq, and Ollama with cost-aware allocation, usage analytics, model governance scoring, and CI regression gating. Built **Leasel** - an 8-agent LangGraph agentic platform with tool calling, durable state, policy enforcement (regulated decisions hard-routed to humans), and a 5-metric eval engine. At Entrata (7+ years, 100K+ users), owned production systems end-to-end in a regulated SaaS platform. Strong across Python, REST APIs, Docker, Kubernetes, AWS, and Azure (OpenAI); architecture standards, technical governance, and cross-functional stakeholder communication. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Agentic AI · Agent Lifecycle Management · LLM Orchestration | AI Governance · Model Evaluation · Policy Enforcement

AI Cost Management · Usage Analytics · FinOps · Multi-Provider Routing | LLM Evaluation · LLM-as-Judge · Golden Datasets · CI Regression

Guardrails · Observability · Production Monitoring | RAG Pipelines · Embeddings · Hybrid Search · Prompt Engineering

Azure (OpenAI · ML · Functions · Foundry) · AWS (Lambda · S3 · SQS · API Gateway) | LangGraph · LangChain · Anthropic Claude · OpenAI

Python · FastAPI · TypeScript · Java · Node.js | Docker · Kubernetes · GitHub Actions · CI/CD | REST APIs · OAuth2 · Secure Authentication · API-Driven Automation

Architecture Standards · Technical Governance · Compliance | Mentoring · Cross-Functional Collaboration · Stakeholder Communication

OpenTelemetry · Prometheus · Grafana · Structured Logging · Tracing

EXPERIENCE

Software Engineer

Jun 2026 - Present

Intuit — Fairfax, Virginia, USA

- **Enterprise AI integration and governance:** Integrate Anthropic Claude into production Python/FastAPI financial services - structured-output schemas, RAG pipeline for retrieval grounding, prompt engineering for consistency, and guardrails with input validation, sensitive-data handling, and fallbacks when model output fails checks.
- **LLM evaluation and model governance:** Build LLM-as-judge evaluation harnesses with golden datasets to catch hallucinations and quality drops before release; implement CI regression gates so AI outputs are scored on every deploy; establish evaluation standards and document patterns for team-wide adoption.
- Build ML data pipelines (scikit-learn, XGBoost); develop React/TypeScript front ends; containerize services with Docker through GitHub Actions CI/CD to AWS with structured logging and CloudWatch monitoring.

Senior Software Engineer

Oct 2020 - Jul 2024

Entrata — Apartment & Affordable Housing SaaS (regulated multi-tenant, 100K+ monthly active users)

- Owned mission-critical compliance data pipelines and backend systems end-to-end - HUD income-limit ETL (180K records, zero compliance violations across 6 years), async RabbitMQ message bus with idempotency guarantees, and PHI-compliant AWS S3 migration (Libsodium encryption, 100% coverage).
- Led Docker microservices redesign (50% scalability improvement, 30% faster deployments); optimized 200+ SQL queries (30% performance gain); Python/Bash automation eliminating 80% of manual operational workload.
- Defined architecture standards and engineering practices across 120+ Agile sprints; mentored 5+ engineers; 300+ code reviews; built internal debugging tooling reducing triage time by 40%.

Junior Software Engineer

Sep 2018 - Oct 2020

Entrata India — Pune

- Built backend modules (Python, PHP, Java, PostgreSQL); delivered large-scale data integration workflows (1.6M records); led AWS S3 cloud migration from legacy storage.

PROJECTS

AgentProbe - AI Cost Management, Governance & LLM Evaluation Platform

2025 | Python, FastAPI, PostgreSQL, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Enterprise-grade LLM evaluation and cost management platform: **cost-aware multi-provider routing** across Anthropic, OpenAI, Azure OpenAI, AWS Bedrock, Groq, and Ollama (routes to cheapest model meeting quality bar); **usage analytics and allocation tracking** per provider and run; **model governance** via 5-metric eval engine (accuracy, hallucination rate, retrieval quality, completeness, compliance); **CI regression harness** that gates releases on eval thresholds; full tracing and structured logging across all agent runs.

Leasel - Agentic AI Platform with Policy Enforcement & HITL Controls

2026 | Python, LangGraph (8-agent), FastAPI, ChromaDB, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production agentic AI platform: **8-agent LangGraph workflow** with tool/function calling, durable state, and conditional routing; **policy enforcement hard-coded** - all regulated compliance decisions route to human staff, not the LLM; **RAG pipeline** with OCR ingestion, hybrid search (TF-IDF + semantic reranking), and citation grounding; **CI evaluation harness** with LLM-as-judge scoring: 93.6% precision, 87.9% F1 across 1,240 test cases.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017